

Welcome to

Network Attacks I – Concepts

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Slides are available as PDF, https://codeberg.org/kramse
academy-2026-day1-network-attacks-I.tex in the repo security-courses

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Goals for today



- Explain common network-attack concepts: botnets and DoS/DDoS
- Explain amplification and reflection attacks
- Explain SYN, UDP and ICMP flooding
- Explain ARP poisoning and man-in-the-middle risks

Plan for today



- Attack overview and botnets
- Denial of Service – DoS and DDoS
- Amplification and reflection
- Flooding techniques across the TCP/IP layers
- ARP poisoning and man-in-the-middle

Time schedule



- **FIXME:** 1) Botnets + DoS/DDoS – XX min
- **FIXME:** 2) Amplification / reflection – XX min
- **FIXME:** 3) Flooding techniques – XX min
- **FIXME:** 4) ARP poisoning / MITM – XX min

Types of attacks – overview



- **FIXME:** where network attacks sit among the attack types

Botnets



- **FIXME:** what a botnet is, how it is built and rented
- **FIXME:** example (e.g. Mirai) – keep it accurate

Denial of Service (DoS)



- **FIXME:** definition and the shop-door analogy

Distributed Denial of Service (DDoS)



- **FIXME:** many sources, scale, impact

Amplification & reflection



- **FIXME:** how reflection works (spoofed source)
- **FIXME:** amplification example (e.g. DNS) – verify factor

TCP/IP refresher – the layers



- **FIXME:** the layers relevant to the attacks below

SYN flood



- **FIXME:** half-open connections, the three-way handshake abuse

UDP flood



- **FIXME:** connectionless flooding

ICMP flood



- **FIXME:** ping flood / smurf (if you cover it)

ARP poisoning & man-in-the-middle



- **FIXME:** how ARP poisoning redirects traffic
- **FIXME:** the MITM risk that follows

Attack landscape



- **FIXME:** current stats / where attacks against DK come from

Summary



- **FIXME:** recap the concepts before the hands-on half